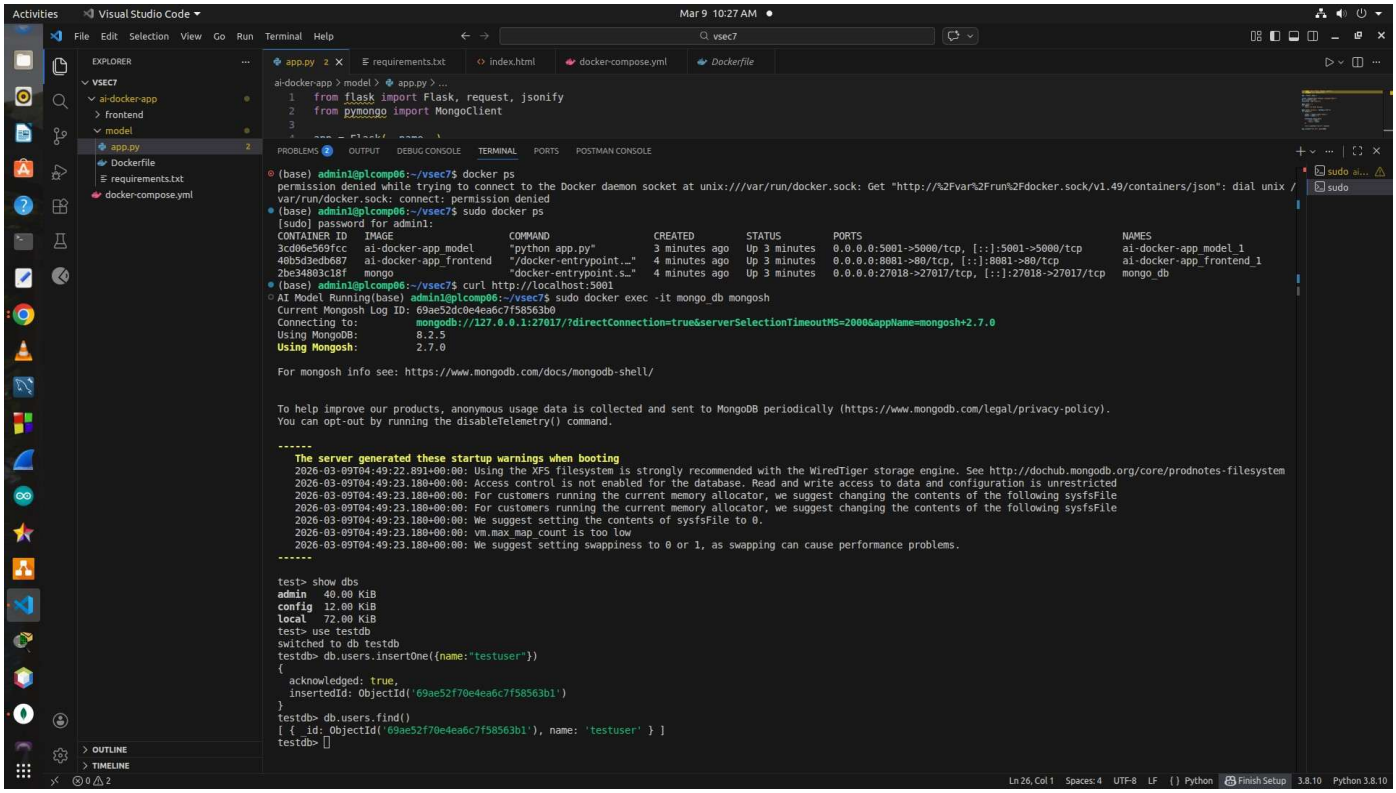
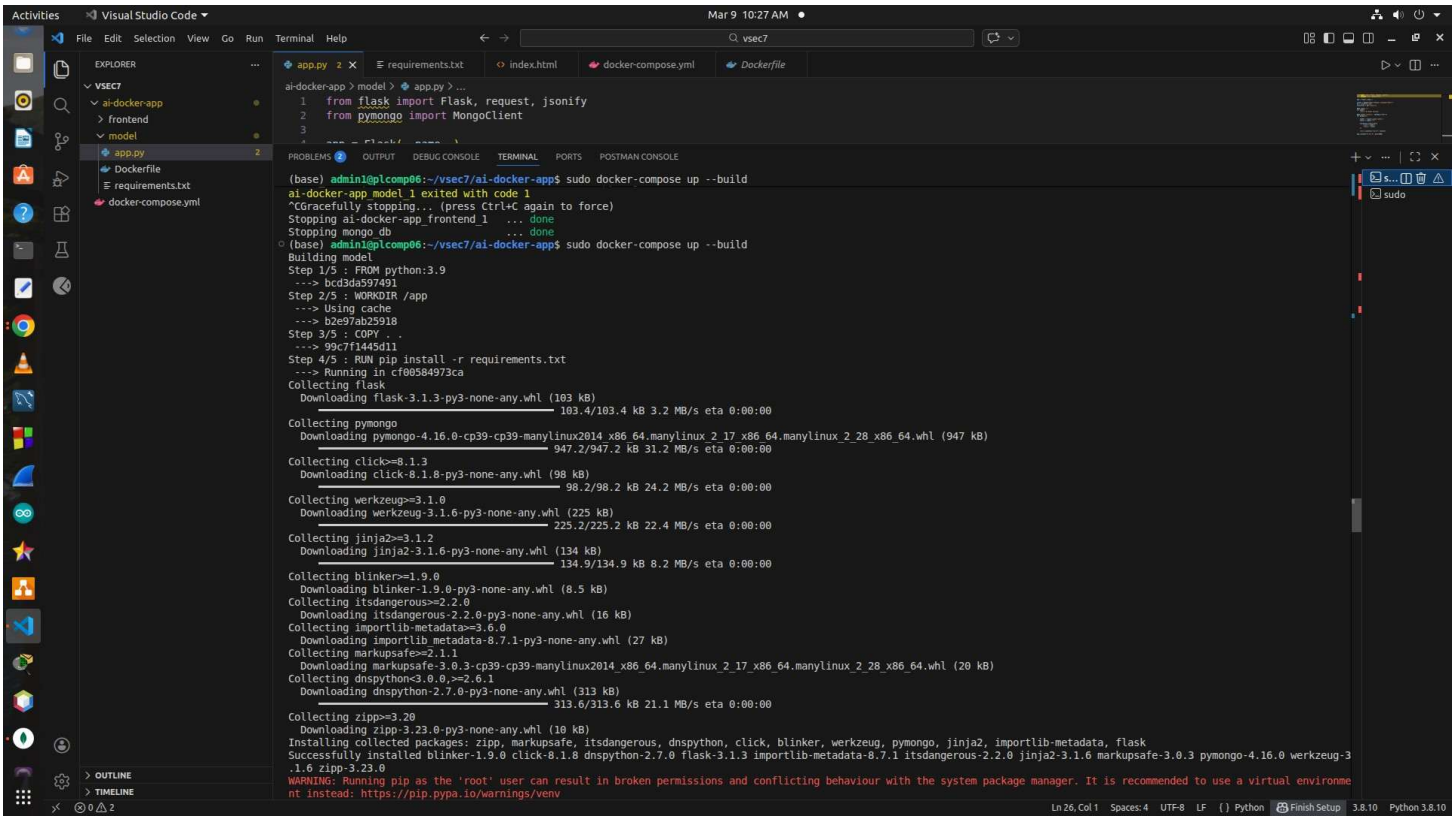
Assignment No.7

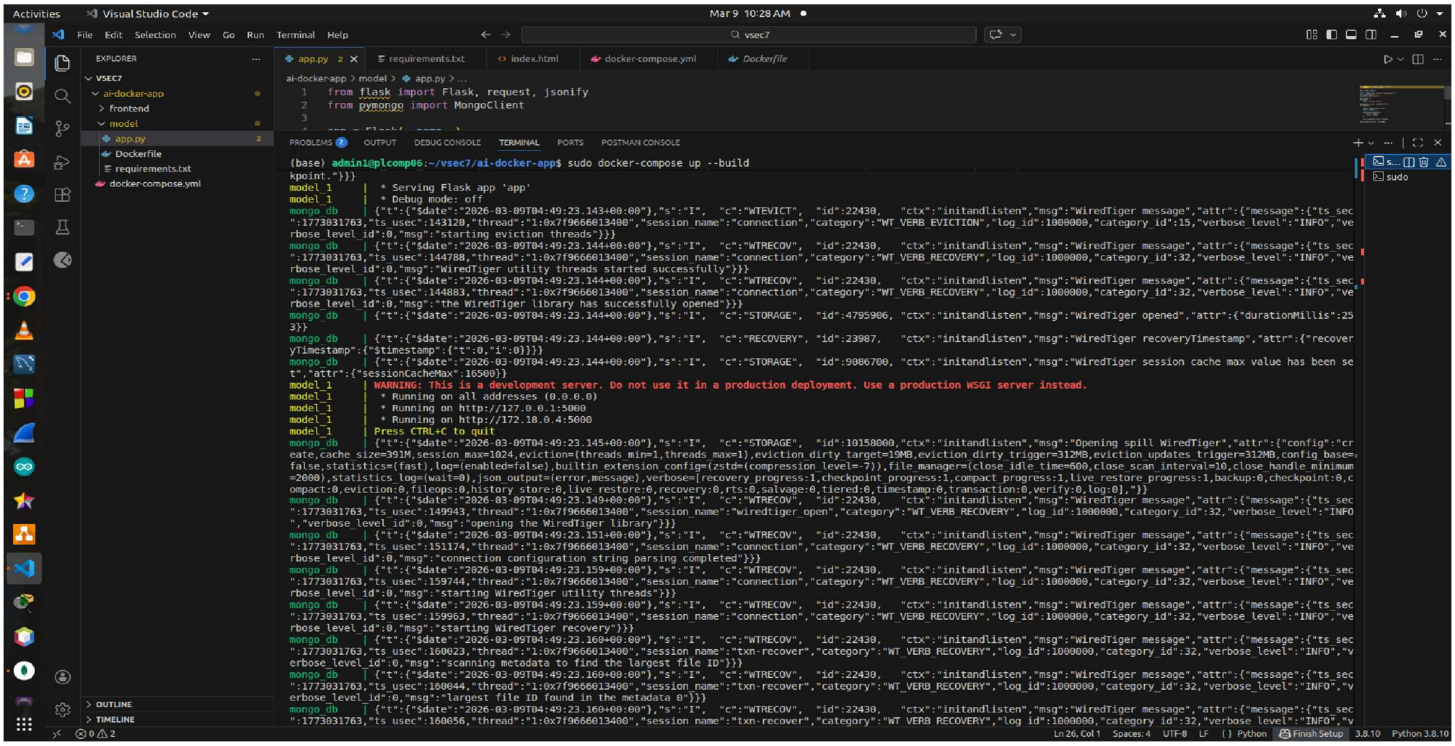
Title: Use Docker Compose to manage multiple containers for a web-based AI app, including containers for the AI model, database, and frontend.

Name: Wavhal Prathmesh Navnath **Div**: TY-B **Roll No:**23107137

Output:




Activities
Visual Studio Code
Mar 9 10:27 AM
File Edit Selection View Go Run Terminal Help
vsec7
EXPLORER
VSEC7
ai-docker-app
frontend
model
app.py
Dockerfile
requirements.txt
docker-compose.yml
app.py 2
requirements.txt
index.html
docker-compose.yml
Dockerfile
ai-docker-app > model > app.py > ...
1 from flask import Flask, request, jsonify
2 from pymongo import MongoClient
3
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE
(base) admin1@plcomp06:~/vsec7/ai-docker-app$ sudo docker-compose up --build
ai-docker-app_model_1 exited with code 1
^CGracefully stopping... (press Ctrl+C again to force)
Stopping ai-docker-app_frontend_1 ... done
Stopping mongo_db ... done
(base) admin1@plcomp06:~/vsec7/ai-docker-app$ sudo docker-compose up --build
Building model
Step 1/5 : FROM python:3.9
---> bcd3da597491
Step 2/5 : WORKDIR /app
---> Using cache
---> b2e97ab25918
Step 3/5 : COPY . .
---> 99c7f1445d11
Step 4/5 : RUN pip install -r requirements.txt
---> Running in cf00584973ca
Collecting flask
Downloading flask-3.1.3-py3-none-any.whl (103 kB)
103.4/103.4 kB 3.2 MB/s eta 0:00:00
Collecting pymongo
Downloading pymongo-4.16.0-cp39-cp39-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (947 kB)
947.2/947.2 kB 31.2 MB/s eta 0:00:00
Collecting click>=8.1.3
Downloading click-8.1.8-py3-none-any.whl (98 kB)
98.2/98.2 kB 24.2 MB/s eta 0:00:00
Collecting werkzeug>=3.1.0
Downloading werkzeug-3.1.6-py3-none-any.whl (225 kB)
225.2/225.2 kB 22.4 MB/s eta 0:00:00
Collecting jinja2>=3.1.2
Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
134.9/134.9 kB 8.2 MB/s eta 0:00:00
Collecting blinker>=1.9.0
Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
Collecting itsdangerous>=2.2.0
Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Collecting importlib-metadata>=3.6.0
Downloading importlib_metadata-8.7.1-py3-none-any.whl (27 kB)
Collecting markupsafe>=2.1.1
Downloading markupsafe-3.0.3-cp39-cp39-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (20 kB)
Collecting dnspython<3.0.0,>=2.6.1
Downloading dnspython-2.7.0-py3-none-any.whl (313 kB)
313.6/313.6 kB 21.1 MB/s eta 0:00:00
Collecting zipp>=3.20
Downloading zipp-3.23.0-py3-none-any.whl (10 kB)
Installing collected packages: zipp, markupsafe, itsdangerous, dnspython, click, blinker, werkzeug, pymongo, jinja2, importlib-metadata, flask
Successfully installed blinker-1.9.0 click-8.1.8 dnspython-2.7.0 flask-3.1.3 importlib-metadata-8.7.1 itsdangerous-2.2.0 jinja2-3.1.6 markupsafe-3.0.3 pymongo-4.16.0 werkzeug-3.1.6 zipp-3.23.0
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv
sudo
OUTLINE
TIMELINE
Ln 26, Col 1 Spaces: 4 UTF-8 LF { } Python Finish Setup 3.8.10 Python 3.8.10


Activities
Visual Studio Code
Mar 9 10:28 AM
File Edit Selection View Go Run Terminal Help
vsec7
EXPLORER
VSEC7
ai-docker-app
frontend
model
app.py
Dockerfile
requirements.txt
docker-compose.yml
app.py 2
requirements.txt
index.html
docker-compose.yml
Dockerfile
ai-docker-app > model > app.py > ...
1 from flask import Flask, request, jsonify
2 from pymongo import MongoClient
3
PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE
(base) admin1@plcomp06:~/vsec7/ai-docker-app$ sudo docker-compose up --build
model_1 | * Serving Flask app 'app'
model_1 | * Debug mode: off
mongo_db | {"t":{"$date":"2026-03-09T04:49:23.143+00:00"},"s":"I", "c":"WTEVICT", "id":22430, "ctx":"initandlisten","msg":"WiredTiger message","attr":{"message":{"ts_sec":1773031763,"ts_usec":143120,"thread":"1:0x7f9666013400","session_name":"connection","category":"WT_VERB_EVICTION","log_id":1000000,"category_id":15,"verbose_level":"INFO","verbose_level_id":0,"msg":"starting eviction threads"}}}
mongo_db | {"t":{"$date":"2026-03-09T04:49:23.144+00:00"},"s":"I", "c":"WTRECOV", "id":22430, "ctx":"initandlisten","msg":"WiredTiger message","attr":{"message":{"ts_sec":1773031763,"ts_usec":144788,"thread":"1:0x7f9666013400","session_name":"connection","category":"WT_VERB_RECOVERY","log_id":1000000,"category_id":32,"verbose_level":"INFO","verbose_level_id":0,"msg":"WiredTiger utility threads started successfully"}}}
mongo_db | {"t":{"$date":"2026-03-09T04:49:23.144+00:00"},"s":"I", "c":"WTRECOV", "id":22430, "ctx":"initandlisten","msg":"WiredTiger message","attr":{"message":{"ts_sec":1773031763,"ts_usec":144883,"thread":"1:0x7f9666013400","session_name":"connection","category":"WT_VERB_RECOVERY","log_id":1000000,"category_id":32,"verbose_level":"INFO","verbose_level_id":0,"msg":"the WiredTiger library has successfully opened"}}}
mongo_db | {"t":{"$date":"2026-03-09T04:49:23.144+00:00"},"s":"I", "c":"STORAGE", "id":4795906, "ctx":"initandlisten","msg":"WiredTiger opened","attr":{"durationMillis":253}}
mongo_db | {"t":{"$date":"2026-03-09T04:49:23.144+00:00"},"s":"I", "c":"RECOVERY", "id":23987, "ctx":"initandlisten","msg":"WiredTiger recoveryTimestamp","attr":{"recoveryTimestamp":{"$timestamp":{"t":0,"i":0}}}}
mongo_db | {"t":{"$date":"2026-03-09T04:49:23.144+00:00"},"s":"I", "c":"STORAGE", "id":9086700, "ctx":"initandlisten","msg":"WiredTiger session cache max value has been set","attr":{"sessionCacheMax":16500}}
model_1 | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
model_1 | * Running on all addresses (0.0.0.0)
model_1 | * Running on http://127.0.0.1:5000
model_1 | * Running on http://172.18.0.4:5000
model_1 | Press CTRL+C to quit
mongo_db | {"t":{"$date":"2026-03-09T04:49:23.145+00:00"},"s":"I", "c":"STORAGE", "id":10158000,"ctx":"initandlisten","msg":"Opening spill WiredTiger","attr":{"config":"create,cache_size=391M,session_max=1024,eviction=(threads_min=1,threads_max=1),eviction_dirty_target=19MB,eviction_dirty_trigger=312MB,eviction_updates_trigger=312MB,config_base=false,statistics=(fast),log=(enabled=false),builtin_extension_config=(zstd=(compression_level=-7)),file_manager=(close_idle_time=600,close_scan_interval=10,close_handle_minimum=2000),statistics_log=(wait=0),json_output=(error,message),verbose=[recovery_progress:1,checkpoint_progress:1,compact_progress:1,live_restore_progress:1,backup:0,checkpoint:0,compact:0,eviction:0,fileops:0,history_store:0,live_restore:0,recovery:0,rts:0,salvage:0,tiered:0,timestamp:0,transaction:0,verify:0,log:0],"}}
mongo_db | {"t":{"$date":"2026-03-09T04:49:23.149+00:00"},"s":"I", "c":"WTRECOV", "id":22430, "ctx":"initandlisten","msg":"WiredTiger message","attr":{"message":{"ts_sec":1773031763,"ts_usec":149943,"thread":"1:0x7f9666013400","session_name":"wiredtiger_open","category":"WT_VERB_RECOVERY","log_id":1000000,"category_id":32,"verbose_level":"INFO","verbose_level_id":0,"msg":"opening the WiredTiger library"}}}
mongo_db | {"t":{"$date":"2026-03-09T04:49:23.151+00:00"},"s":"I", "c":"WTRECOV", "id":22430, "ctx":"initandlisten","msg":"WiredTiger message","attr":{"message":{"ts_sec":1773031763,"ts_usec":151174,"thread":"1:0x7f9666013400","session_name":"connection","category":"WT_VERB_RECOVERY","log_id":1000000,"category_id":32,"verbose_level":"INFO","verbose_level_id":0,"msg":"connection configuration string parsing completed"}}}
mongo_db | {"t":{"$date":"2026-03-09T04:49:23.159+00:00"},"s":"I", "c":"WTRECOV", "id":22430, "ctx":"initandlisten","msg":"WiredTiger message","attr":{"message":{"ts_sec":1773031763,"ts_usec":159744,"thread":"1:0x7f9666013400","session_name":"connection","category":"WT_VERB_RECOVERY","log_id":1000000,"category_id":32,"verbose_level":"INFO","verbose_level_id":0,"msg":"starting WiredTiger utility threads"}}}
mongo_db | {"t":{"$date":"2026-03-09T04:49:23.159+00:00"},"s":"I", "c":"WTRECOV", "id":22430, "ctx":"initandlisten","msg":"WiredTiger message","attr":{"message":{"ts_sec":1773031763,"ts_usec":159963,"thread":"1:0x7f9666013400","session_name":"connection","category":"WT_VERB_RECOVERY","log_id":1000000,"category_id":32,"verbose_level":"INFO","verbose_level_id":0,"msg":"starting WiredTiger recovery"}}}
mongo_db | {"t":{"$date":"2026-03-09T04:49:23.160+00:00"},"s":"I", "c":"WTRECOV", "id":22430, "ctx":"initandlisten","msg":"WiredTiger message","attr":{"message":{"ts_sec":1773031763,"ts_usec":160023,"thread":"1:0x7f9666013400","session_name":"txn-recover","category":"WT_VERB_RECOVERY","log_id":1000000,"category_id":32,"verbose_level":"INFO","verbose_level_id":0,"msg":"scanning metadata to find the largest file ID"}}}
mongo_db | {"t":{"$date":"2026-03-09T04:49:23.160+00:00"},"s":"I", "c":"WTRECOV", "id":22430, "ctx":"initandlisten","msg":"WiredTiger message","attr":{"message":{"ts_sec":1773031763,"ts_usec":160044,"thread":"1:0x7f9666013400","session_name":"txn-recover","category":"WT_VERB_RECOVERY","log_id":1000000,"category_id":32,"verbose_level":"INFO","verbose_level_id":0,"msg":"largest file ID found in the metadata 0"}}}
mongo_db | {"t":{"$date":"2026-03-09T04:49:23.160+00:00"},"s":"I", "c":"WTRECOV", "id":22430, "ctx":"initandlisten","msg":"WiredTiger message","attr":{"message":{"ts_sec":1773031763,"ts_usec":160056,"thread":"1:0x7f9666013400","session_name":"txn-recover","category":"WT_VERB_RECOVERY","log_id":1000000,"category_id":32,"verbose_level":"INFO","v
sudo
OUTLINE
TIMELINE
Ln 26, Col 1 Spaces: 4 UTF-8 LF { } Python Finish Setup 3.8.10 Python 3.8.10